

Information Theory

Review of Probability Theory

Consider an experiment X with possible outcomes in a finite set $\mathcal{X}$. For example X could be flipping a coin and $\mathcal{X} = \{\text{heads}, \text{tails}\}$.

each outcome is assigned a probability.

$$P_1 = p(X=\text{head}) = \frac{1}{2}, \quad p(X=\text{tail}) = \frac{1}{2}.$$

Often outcome X of an experiment is called a random variable.

In general $\forall x \in \mathcal{X}$, denote the probability $X=x$ by

$$p_X(x) = p_x = p(X=x)$$

Note that $\sum_{x \in \mathcal{X}} p_x = 1$

If $A \subseteq \mathcal{X}$ then let

$p(A) = \sum_{x \in A} p_x$ which is the probability that x takes a value in A

Often one performs an experiment where one is measuring several different events. These events may or may not be related but they may be dumped together to form a new random event

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If we have 2 random variables X & Y with possible outcomes in X, Y respectively, then we may create a new random event $Z = (X, Y)$
The new event Z has a set of possible outcomes

$Z = X \times Y$. Z is Joint Random Variable

Example:

~~Draw~~ Draw a ~~card~~ card from a standard deck

Let X be suit of card .

$X = \{ \text{clubs, hearts, diamonds, spades} \}$

Let Y be ~~the~~ value of card no $Y = \{ 2, 3, \dots, ace \}$

Then Z gives 52 possibilities - for card. Note $x \in X$
 $y \in Y$, then $p((X, Y) = (x, y)) = p(X=x, Y=y)$

is simply the prob. that card drawn has suit x and value y .

Since all cards are equally probable, the probability is $\left(\frac{1}{52}\right)$ which is the probability that $X=x$ (namely $1/4$) $\left(\frac{1}{13}\right)$

Times the probability $Y=y$ (namely $1/13$) . This means

X & Y are independent.

Ex Roll a die. Suppose you are interested in 2 things whether the no of dots ~~is~~ is odd and whether it is at least 2. Let $X=0$ if no of dots is even and $X=1$ if no of dots is odd.

Let $Y=0$ if no of dots < 2 .

$Y=1$ if no of dots is at least 2. (~~is~~ ≥ 2)

Then $Z=(X,Y)$ gives result of both experiments together. Note that $P(Z=(1,0)) = \frac{1}{6} \neq P(X=1) \times P(Y=0)$

~~$\neq P(X=1) \times P(Y=0)$~~

$\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

This means X and Y are not independent. This is closely related to fact that knowing X gives us information about Y .

rule denote $P_{X,Y}(x,y) = P(X=x, Y=y)$

We can recover the probability $X=x$.

as $P_X(x) = \sum_{y \in Y} P_{X,Y}(x,y) \quad \forall x \in X$

Conditional Probability

We are interested in probability of Y given $X=x$ has occurred if $P_X(x) > 0$ defines conditional probability of $Y=y$ given $X=x$

$$p_Y(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$$

$\therefore$ Total probability $p_X(x) = \sum_y p_{X,Y}(x,y)$. The fraction of this sum that comes from $Y=y$ is $p_Y(y|x)$

Note that X & Y are independent iff $p_Y(y|x) = p_Y(y)$
 $\forall x, y$

Bayes Th: If $p_X(x) > 0$ and $p_Y(y) > 0$ then

$$p_X(x|y) = \frac{p_X(x) \cdot p_Y(y|x)}{p_Y(y)}$$

ENTROPY

Roll a 6-sided die and a 10-sided die. which experiment has more uncertainty?

If you make a guess of outcome of each roll, you are more likely to be wrong with 10-sided die.

Therefore 10-sided die has more uncertainty.
 Similarly, compare a fair-coin toss in which heads and tails are equally likely with a coin toss in which heads occur 90% of the time, which has more uncertainty.
 Fair coin toss has more uncertainty.

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In our defⁿ of ~~cert~~ uncertainty, we want to make sure
2 random variabls. X & Y that have same probability
distribⁿ has same uncertainty.

We require measure of uncertainty to satisfy the
following properties:

(1) For each set of non-ve nos $p_1, p_2, \dots, p_n$ with $p_1 + p_2 + \dots + p_n = 1$

the uncertainty is given by $H(p_1, p_2, \dots, p_n)$
~~continuous~~

(2) H should be a continuous function of probability
distribution, so a small change in prob distribⁿ
should not drastically change the ~~cert~~ uncertainty.

(3) $H\left(\frac{1}{n+1}, \frac{1}{n+1}, \dots\right) \leq H\left(\frac{1}{n+1}, \frac{1}{n+1}, \dots, \frac{1}{n+1}\right)$

$\forall n \geq 0$. In other words, in situation where
all outcomes are equally likely, the uncertainty
increases when there are more possible outcomes.

(4) If $0 < q < 1$ then $H(p_1, \dots, q, p_j, (1-q)p_j, \dots, p_n)$
 $= H(p_1, \dots, p_j, \dots, p_n) + p_j H(q, 1-q)$

Entropy

(4) If $0 < q < 1$ then

$$H(p_1, \dots, q/p_j, (1-q)p_j, \dots, p_n) = H(p_1, \dots, p_n) + p_j H(q, 1-q)$$

This means j^{th} outcome is broken into 2 sub-outcomes with probabilities q/p_j and $(1-q)p_j$, then total uncertainty is increased by uncertainty caused by choice between 2 sub-outcomes multiplied by probability p_j .

eg

If we roll a 6-sided die, we can record 2 outcomes: even and odd. This has uncertainty $H(\frac{1}{2}, \frac{1}{2})$. Now suppose we break the outcome even into 2 sub-outcomes

{2, 4} and {4, 6}. Then we have 3 possibilities

2, {4, 6} and odd, we have, $H(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}) = H(\frac{1}{2}, \frac{1}{2}) + \frac{1}{2} H(\frac{2}{3}, \frac{1}{3})$

1st term is the uncertainty caused by even ~~or~~ or odd. The 2nd term is the uncertainty added by splitting even into 2 sub-outcomes

Shannon Entropy

Thm: Let $H(X)$ be a function satisfying properties (1)-(4). In other words: for each random variable X , with outcomes $X = \{x_1, \dots, x_n\}$ having probabilities $p_1, \dots, p_n$

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the fⁿ H assigns a no $H(X)$ subject to conditions (1) - (4)

Then H must be of form

$$H(p_1, \dots, p_n) = - \sum_k p_k \log p_k$$

(where n is non-ve count & where the sum is taken over those k such that $p_k > 0$)

Because of theorem we define entropy of the variable X to be $H(X) = - \sum_{x \in X} p(x) \log_2 p(x)$

$H(X)$ is measure of uncertainty in outcomes X .

$$\text{Since } 0 < p(x) \leq 1$$

$$\Rightarrow \log_2 p(x) \leq 0$$

$\Rightarrow H(X) \geq 0$, so there is no such thing as -ve ^{uncertainty} ~~uncertainty~~

There are problems when we have elements $x \in X$ that have probability 0.

In this case, we define $0 \log_2 0 = 0$

$$\lim_{x \rightarrow 0} x \log x = 0$$

Entropy of X may also be interpreted as the expected value of $-\log_2 p(x)$ [recall $E[g(x)] = \sum_x g(x) p(x)$]

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Example:

Consider a fair coin toss. There are 2 outcomes, each with probability $\frac{1}{2}$. The entropy

$$-\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}\right) = 1 \Rightarrow \text{Result of coin flip gives 1 bit of information.}$$

Ex: Consider a non fair coin: toss X with probability p getting heads & probability $(1-p)$ getting tails ($0 < p < 1$). The entropy of event $H(X) = -p \log_2 p - (1-p) \log_2 (1-p)$.

If one considers $H(X)$ as a function of p , an

$$\begin{aligned} \frac{dH(X)}{dp} &= -\log_2 p - 1 + 1 + \log_2 (1-p) = 0 \\ &= \log_2 \left(\frac{1-p}{p} \right) = 0 \end{aligned}$$

$\Rightarrow$

$p = 1/5$	$p = 1/2$
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Example:

Consider a n sided fair die. There are n outcomes each with probability $\frac{1}{n}$.

Entropy is

$$\begin{aligned} & -\frac{1}{n} \log_2 \left(\frac{1}{n} \right) \cdots - \left(\frac{1}{n} \right) \log_2 \left(\frac{1}{n} \right) \\ &= \boxed{\log_2 n} \end{aligned}$$

There is a relationship b/w ENTROPY and the number of YES-NO questions needed to determine an outcome.

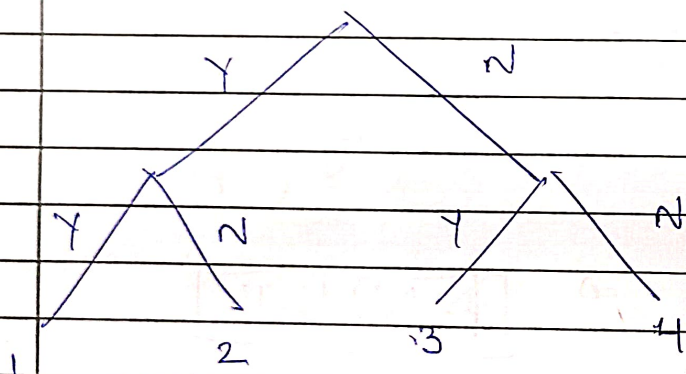
If one considers a ~~totally~~ ^{totally} non-fair coin tosses with $p(1) = 1$.

$$H(X) = -1 \log_2 1 - 0 \cdot \log_2 0 = 0$$

$\Rightarrow$ NO QUESTION

If you roll a 4-sided die then it takes 2 questions to determine the outcome. $\log_2 4 = 2$ questions

(g) Is the no less than 3. $\rightarrow$ g1



Is the no odd $\rightarrow$ g2

A slightly more subtle example is obtained by flipping 2 coins. Let X be no of heads, so possible outcomes are $\{0, 1, 2\}$. The probabilities are $\{1/4, 1/2, 1/4\}$

$$\text{Entropy is } -\frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{4} \log_2 \frac{1}{4}$$

TT $\rightarrow$ 0

TH $\rightarrow$ 1

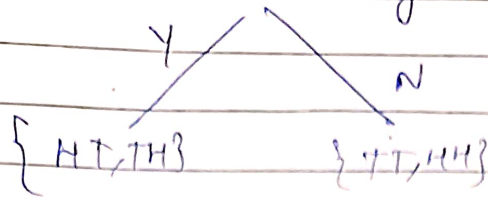
HT $\rightarrow$ 1

HH $\rightarrow$ 2

$$= \boxed{\frac{3}{2}}$$

We can on an average ask $\frac{3}{2}$ questions

For eg $\rightarrow$ Q1 $\rightarrow$ Is there exactly 1 head?



Q2 $\rightarrow$ Are there 2 heads?

~~$\frac{1}{2} \binom{1}{2}$~~

Expected $g = \binom{1}{2} (1) + \binom{1}{2} (2) = \boxed{\frac{3}{2}}$

Q3

hyperledger

virtual machine monitor $\rightarrow$ Hypervisor

PowerPC $\rightarrow$ x86 $\rightarrow$ ARM

for mac

Posetta
Posetta gk

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Entropy

Another way of looking at entropy $H(X)$ is that it measures the no. of bits of information that we obtain when we are given the outcome of X . in a 4bit random number where each ~~probab~~ possibility has probability $1/16$, Entropy $H(X) = 4$ which says we have received 4 bits of info.

In a similar vein entropy relates to minimal amount of bits necessary to represent an event on a computer (which is a binary device). There is no sense recording events whose outcome can be predicted with 100% certainty, it would be waste of space. In storing information one just wants to code the uncertain parts because there is where real implementation is.

Joint Entropy

If we have 2 random variables X and Y . the joint entropy $H(X, Y)$ is defined as

$$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p_{X,Y}(x, y) \log_2 p_{X,Y}(x, y)$$

This is just entropy of random variable $Z = (X, Y)$

In a cryptosystem, we might want to know the uncertainty of a given key given the knowledge of ciphertext. This includes us to concept of conditional entropy which is the amount of uncertainty

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in Y ; given X

$$\begin{aligned} H(Y/X) &= \sum_x p_X(x) H(Y/X=x) \\ &= - \sum_x p_X(x) \sum_y p_Y(y/x) \log_2 p_Y(y/x) \\ &= - \sum_x \sum_y p_{XY}(x,y) \log_2 p_Y(y/x) \end{aligned}$$

The last equality follows from relationship

$$p_{XY}(x,y) = p_Y(y/x) p_X(x)$$

The quantity $H(Y/X=x)$ is uncertainty of Y given $X=x$. It is defined in terms of conditional probabilities by expression within parenthesis in 2nd line.

We calculate $H(Y/X)$ by forming a weighted sum of ~~these~~ these uncertainties to get total uncertainty in Y given that we know value of X .

Remark:

The preceding ~~a~~ defⁿ of conditional entropy uses weighted avg over various ~~x~~ x of entropy of Y given $X=x$.

Math: $H(Y/X) = - \sum_{x,y} p_{XY}(x,y) \log_2 p_Y(y/x)$

If X and Y are independent

$$H(X,Y) = H(X) + H(Y) = H(Y/X)$$

for example if X and Y are independent, then the defⁿ would imply the uncertainty of Y given X is greater than uncertainty of Y

$$H(Y|X) > H(Y)$$

It should be $H(Y|X) = H(Y)$ when X & Y are independent

we now derive an important tool,
the chain rule for entropy

thm (Chain Rule) $H(X, Y) = H(X) + H(Y|X)$

$$\text{pf: } H(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X, Y}(x, y) \log_2 p_{X, Y}(x, y)$$

$$= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X, Y}(x, y) \log_2 p_X(x) p_{Y|X}(y|x)$$

$$= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X, Y}(x, y) \log_2 p_X(x)$$

$$- \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X, Y}(x, y) \log_2 p_{Y|X}(y|x)$$

$$= - \left(\sum_x \log_2 p_X(x) \sum_y p_{X, Y}(x, y) \right) + H(Y|X)$$

$$= - \sum_x p_X(x) \log_2 p_X(x) + H(Y|X)$$

$$= H(X) + H(Y|X)$$

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What does chain rule tell us?

The uncertainty of joint ~~entropy~~^{event} (X, Y) is equal to a
uncertainty of event X + uncertainty of event Y .
given that event X has happened

Huffman Code

Relationship b/w Entropy & Compression is first clear

more Compression $\rightarrow$ Higher Randomness $\rightarrow$ Higher Entropy

Ex: Suppose we have an alphabet

with uniform length coding to encode 4 letters we need 2 bits

Letter	Code
a	00
b	01
c	10
d	11

What about variable length encoding?

Morse Code

Variable Length Code

more frequent letters were provided with smaller representation

A-B-C

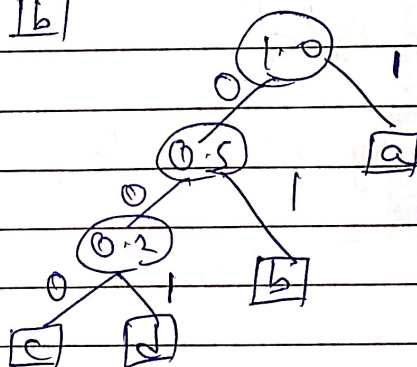
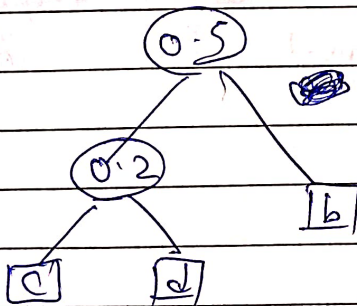
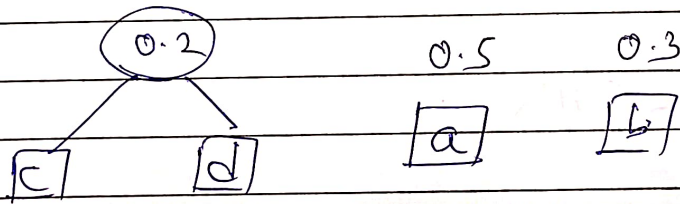
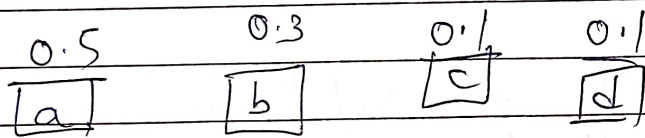
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Letter	Code
l	0
t	1
re	100
z	1100

Variable length code, should be unique prefix code i.e no code should be proper prefix of other.

This enables in unambiguous decoding

Huffman Code



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Letter	Huffman Code
a	1
b	01
c	000
d	001

$$\text{Avg Code length} = 1 \times 0.5 + (2) \times 0.3 + 3 \times 0.1 + 3 \times 0.1$$

$$= 0.5 + 0.6 + 0.3 + 0.3$$

$$= \boxed{1.7} < 2$$

The Average no of bits per output is closely related to entropy

Thm: Let L be avg no of bits per output for Huffman Encoding for random variable X .

Then

$$H(X) \leq L < H(X) + 1$$

The result agrees with interpretation that entropy measures how many bits of information is contained with output of X .

In ~~one~~ ^{our} ~~example~~ entropy is

$$H(X) = -(0.5 \log_2 0.5 + 0.3 \log_2 0.3 + 0.1 \log_2 0.1 + 0.1 \log_2 0.1)$$

$$\approx 1.68 \text{ (very close to } 1.7)$$

Secretary Perfect Secrecy

Suppose we have a cipher system with possible plaintext P , ciphertext C and keys K . Each plaintext in P has certain probability of occurring some are more likely than others. The choice of a key in K is always assumed to be independent of choice of plaintext. The possible ciphertexts in C have various probabilities for P and K . If Eve intercept a ciphertext, how much information does she obtain for key? In other words what $H(K|C)$? Initially the uncertainty $H(K)$ has knowledge of ciphertext decreased the uncertainty

Ex: Suppose we have 3 possible plain text: a, b, c with probabilities $0.5, 0.3, 0.2$ and 2 keys k_1 and k_2 with probabilities 0.5 and 0.5 . Suppose possible ciphertexts are u, v, w . Let e_k be the encryption f^* for key k . Suppose

$$e_{k_1}(a) = u, e_{k_1}(b) = v, e_{k_1}(c) = w$$

$$e_{k_2}(a) = v, e_{k_2}(b) = w, e_{k_2}(c) = u$$

Let $p_p(a)$ denote probability of plaintext is a . probability that ciphertext is u is

$$p_c(u) = p_k(k_1) p_p(a) + p_k(k_2) p_p(c)$$

$$= (0.5)(0.5) + (0.5)(0.2)$$

$$= 0.35$$

$$p_c(v) = p_k(k_1) p_p(b) + p_k(k_2) p_p(a)$$

$$= 0.5 \times 0.3 + 0.5 \times 0.5 = 0.15 + 0.25 = 0.4$$

$$\text{Similarly } p_c(w) = 0.25,$$

Suppose someone intercepts a ciphertext. This gives some information on plain text: for example if ciphertext is U then it can be deduced ~~is~~ immediately plaintext is a. If ciphertext is V then plaintext can be b or c

$$p(b|v) = \frac{p(b, v)}{p_c(v)}$$

$$= \frac{p(b, v)}{p_c(v)} = \frac{0.3 \times 0.5}{0.25} = \boxed{0.6}$$

$$p(c|v) = 0.4, \quad p(a|v) = 0$$

Similarly:

$$p(a|w) = 0, \quad p(b|w) = 0.6, \quad p(c|w) = \underline{0.9}$$

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Perfect Secrecy

Ex

$$p_P(a) = 0.5$$

$$p_P(b) = 0.3$$

$$p_P(c) = 0.2$$

$$p(a|v) = 0 \quad p(a|w) = 0$$

$$p(b|v) = 0.6 \quad p(b|w) = 0$$

$$p(c|v) = 0.4 \quad p(c|w) = 0$$

Knowledge of ciphertext allows us to zero probabilities

Therefore ciphertext gives us information about plaintext
we can quantify this via concept of conditional entropy

Entropy of plaintext:

$$H(P) = -(0.5 \log_2 0.5 + 0.3 \log_2 0.3 + 0.2 \log_2 0.2) = 1.785$$

conditional entropy of P given C is

$$E(P|C) = - \sum_{x \in \{a,b,c\}} \sum_{y \in \{v,w\}} p(x) p(y|x) \log_2 p(y|x)$$

$$= 0.485$$

The uncertainty of plain text $\downarrow$ when cipher text is known

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Defⁿ: A cryptosystem has perfect secrecy.
if $H(K|C) = H(K)$.

Conditional Entropy $H(K|C)$ is called key equivocation, it is a measure of amount of uncertainty of key remaining when ciphertext is known.

Theorem: Let (P, C, K, E, D) be a cryptosystem.

$$\text{Then } H(K|C) = H(K) + H(P) - H(C)$$

pf: First observe that $H(K, P, C)$
 $= H(C|K, P) + H(K, P)$

The key & the plaintext determines ciphertext uniquely since $y = E_K(x)$

This implies $H(C|K, P) = 0$
 Hence $H(K, P, C) = H(K, P)$

But K and P are independent
 $\therefore H(K, P) = H(K) + H(P)$

Hence $H(K, P, C) = H(K, P) = H(K) + H(P)$

- In a similar fashion, since key and ciphertext determine plaintext uniquely i.e. $x = D_K(y)$, we have $H(P|K, C) = 0$
 Hence $H(K, P, C) = H(K, C)$
 Now we compare as follows:

$$\begin{aligned}
 H(K|C) &= H(K, C) - H(C) \\
 &= H(K, C) - H(C) \\
 &= H(K) + H(C) - H(C)
 \end{aligned}$$

Thm:

Consider a cryptosystem such that

- (1) Every key has probability $1/\#K$ [$\#K$ denotes the no. of possible keys]
- (2) For each $n \in P$ and $c \in C$ there is exactly one key $k \in K$ such that $E_k(n) = c$

Then this cryptosystem has Perfect secrecy.

ONE TIME PAD (Perfect Secrecy)

It is easy to deduce condition 2 so that $\#C = \#K$. Conversely it can be shown that if $\#P = \#C = \#K$ and the system has perfect secrecy, then (1) & (2) hold.

It is natural to ask how preceding concept apply to a cryptosystem. A possibly surprising answer is that $H(P|C) = 0$. namely the ciphertext determines plaintext. Reason is that ERROR doesn't take into account computation time. What counts is that all the information needed to recover plaintext is contained in knowledge of n , e and c .

Spurious keys & Uncertainty Distance

Letter (P, C, E, D) is the cryptosystem being used

A string of plaintext

$x_1, x_2, \dots, x_n$

- is encrypted with one key producing a string of ciphertext $y_1, y_2, \dots, y_n$

Goal of cryptanalyst is to determine key

We are looking at ciphertext only attacks and we assume infinite computational resources. We also assume that we know plaintext is natural lang such as ENGLISH. Oscar will be able to rule out certain keys but many possible keys may remain. One of which is correct. The remaining possible keys are called spurious keys.